

YANG YANG (4TH YEAR PHD CANDIDATE)

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Keywords: Computer Architecture/GPU/Confidential Computing/Cryptography/Performance/Memory Technology

I work on **PERFORMANCE** issues in **GPU** × **CONFIDENTIAL COMPUTING** × **CRYPTOGRAPHY** × **MEMORY**

Recently, I have focused on performance issues in both conventional GPU environments (e.g., multi-GPU systems [ISCA'25]) and secure GPU environments such as confidential computing (e.g., TDX-based H100 systems [ISPASS'25, ISCA'26]).

My research interests span (a) *scaling GPU CC*; (b) *addressing the memory wall in GPU CC*; (c) *designing GPU/CC-friendly cryptographic protocols (FHE/PIR)*; and (d) *accelerating privacy-preserving LLM inference*.

EDUCATION

 **UNIVERSITY OF VIRGINIA, Charlottesville, USA**

AUG. 2023 - NOW

 **PH.D. in Computer Science**

GPA: 3.91/4.0

Topics: GPU × {Trusted-Computing, Cryptography, Memory}

Advisor: Dr. Adwait Jog

 **JILIN UNIVERSITY, Changchun, China**

SEPT. 2019 - JUL. 2023

 **B.S. in Computer Science**

GPA: 3.69/4.0

Thesis: The Design and Implementation of Binary Code Analysis Framework for NVIDIA GPU.

Advisor: Dr. Jingweijia Tan

PUBLICATIONS

[1] (ISCA'26)

LEGIS: Pinpointing and Addressing Performance Overheads of GPU-based Confidential Computing

YANG YANG, Adwait Jog

In the Proceedings of IEEE/ACM International Symposium on Computer Architecture, Raleigh, NC, June 2026

[2] (ISCA'25)

NETCRAFTER: Tailoring Network Traffic for Non-Uniform Bandwidth Multi-GPU Systems

Amel Fatima, YANG YANG, Yifan Sun, Rachata Ausavarungnirun, Adwait Jog

In the Proceedings of ACM International Symposium on Computer Architecture, Tokyo, Japan, June 2025

[3] (ISPASS'25)

Dissecting Performance Overheads of Confidential Computing on GPU-based Systems

YANG YANG, Mohammad Sonji, Adwait Jog

In the Proceedings of IEEE International Symposium on Performance Analysis of Systems and Software, Ghent, Belgium, May 2025

[4] (SRC@CGO'23) [poster, before-PhD]

Facilitating Profile Guided Compiler Optimization with Machine Learning

YANG YANG, Xueying Wang, Guangli Li

SRC of IEEE/ACM International Symposium on Code Generation and Optimization, Montreal, Canada, February 2023

EXPERIENCE

GPU Hardware Security Intern

MAY. 2026 - NOW

NVIDIA Corporation, Santa Clara, California, USA

Manager: Vidhya Krishnan, Ambarish Vyas **Mentor:** Santosh Ghosh, Moein Ghaniyoun

Collaborators: Michael Sullivan (**ARG**), Chaz Gouert, Jean-Luc Watson, Edward Suh (**Security & Privacy Research**)

Focus: HEDGE: Homomorphic Encryption–Augmented GPU Confidential Computing for Secure LLM Inference

- Investigated CC support for next-generation NVIDIA GPUs.
- CKKS-encrypted weights for Qwen3 (Dense/MoE/LoRA)/DeepSeek-v4 (MoE).
- Designed a new protocol (OTMHE) with a formal security proof against known attacks.
- Optimized the CKKS on H100/B200, reducing overhead relative to plaintext from $11,500\times$ (SOTA GPU library) to $1.8\times$.
- Developed CKKS execution paths for batched decoding. Achieved $1.18\times$ overhead for Qwen3-32B-512Token-batched.
- Based on internship project: co-inventor on three internal inventions; working on a paper draft with NV Research.

Insight Computer Architecture Lab
University of Virginia, Charlottesville, Virginia, USA
Advisor: Dr. Adwait Jog
Focus: Architectural Support for Efficient and Secure GPU Computing.

AUG. 2023 – NOW

- GPU-centric memory and storage system (DDR/HBM/SSD with UVM via PCIe/CXL/RDMA). [ISCA'25, ISCA'26]
- TDX-based confidential computing on GPUs. [ISPASS'25, ISCA'26]
- Cryptography (FHE/PIR) with GPU acceleration (work closely with **Prof. Wei-Kai Lin**).
- Efficient encryption for GPU-based heterogeneous memory system. [ISCA'26]

State Key Laboratory of Processor
Institute of Computing Technology, Chinese Academy of Science, Beijing, China
Advisor: Dr. Guangli Li / Dr. Xueying Wang
Topics: Compiler & Programming Systems
Focus: Facilitating Profile-Guided Compiler Optimization with Graph Learning

JUL. 2022 – SEPT. 2023

- Proposed a branch predictor using XGBoost based on compile time static analysis. [CGO-SRC'23]
- Utilize graph representations to build predictive profile-guided optimization framework and integrate it into LLVM.

Emerging Technology Enabled Computer Architecture Lab
Jilin University, Changchun, Jilin, China
Advisor: Dr. Jingweijia Tan
Topics: GPU $\times$ {PTX/SASS, Reliability, Energy Efficiency}
Focus: Reliability and Energy-Efficiency of GPUs

FEB. 2022 – JUL. 2023

- Soft error and hard error (e.g., process variation, NBTI) on FinFET based chiplet GPUs.
- SASS level analysing and programming framework for NVIDIA Ampere GPUs. [THESIS]
- Deep learning techniques for GPU power modeling.
- Explored how to reverse-engineer SASS and program in SASS.
- Instruction level under-voltage reliability of GPUs

TALKS

HEDGE: Homomorphic Encryption–Augmented GPU Confidential Computing for Secure LLM Inference

- NVIDIA GPU Hardware Security. July 2026
- NVIDIA Architecture Security. August 2026
- NVIDIA cross team intern project presentation. August 2026

LÆGIS: Pinpointing and Addressing Performance Overheads of GPU-based Confidential Computing

- UVA CS SIG. May 2026
- NVIDIA GPU Hardware Security/NVIDIA Research. June 2026
- IEEE/ACM ISCA (Raleigh). June 2026

NETCRAFTER: Tailoring Network Traffic for Non-Uniform Bandwidth Multi-GPU Systems

- ACM ISCA (Tokyo). June 2025
- UVA CS SIG. September 2025

Dissecting Performance Overheads of Confidential Computing on GPU-based Systems

- IEEE ISPASS (Belgium). May 2025
- UVA CS SIG. May 2025
- IBM Research (Virtual). May 2025
- Confidential Computing Summit. June 2025
- NVIDIA Hardware Security/NVIDIA Research. June 2026

TEACHING

25 FALL / 26 FALL @ UVA, TA for CS 4444: Introduction to Parallel Computing (undergraduate)

- Co-designed CUDA programming assignments.
- Delivered a lecture on GPU-based confidential computing.

24 FALL @ UVA, TA for CS 6354: Computer Architecture (graduate)

- Designed a pipelined RISC-V simulator in Python.

AWARD

[**Division Champion & 3rd Place**] 2026 UVA McIntire Hedge Tournament, with John Farrell (lead) and Lars Christensen

[**Travel Grant**] ISPASS'25 ISCA'25 ISCA'26

[**Fellowship@UVA**] 2023

[**Scholarship@JLU**] 2019 2020 2021 2022

SERVICE

[**Reviewer**] [IEEE Transactions on Computers](#)

[**PC**] [HASP'26](#)

[**Submission Chair**] [IISWC'26](#)

[**GPU-RIG**] Coordinating logistics for [GPU Research Interest Group](#) across [UVA SEAS](#) in 2026/2027.

[**AE Committee**] [IISWC'25](#) [ASPLOS'27](#)

[**SIG**] Co-organizer of Systems-Interest-Group (SIG) meetings at [UVA Department of Computer Science](#) in 2024/2025.

SKILLS

[**Agents**] Claude Code, Codex

[**Languages**] C/C++, Python, Assembly, Go, Rust

[**Frameworks**] CUDA, Pytorch, XGBoost, LLVM, TDX/SGX, QEMU/KVM, NVIDIA GPU Kernel Module, OpenFHE, Cheddar

[**Software**] Linux, ~~La~~TeX, GNU compilers, GPGPU-Sim/Accel-Sim, NVCC/NCU/NSYS/NVBit, DRAMSim, Varius-TC, Z3

[**Cryptography**] AES (GCM, SIV, *etc.*), Merkle Trees, Public-Key (DH, RSA, *etc.*), ORAM, FHE, PIR (LWE/FHE-based)

PHD COMMITTEE (QUAL EXAM)

Chair: [Wei-Kai Lin](#) **Members:** [Sandhya Dwarkadas](#) [Felix Lin](#) [Adwait Jog](#) **Announcement:** [\[Event link\]](#)